Software Engineering Lab 3

Component Modelling & Architecture Selection

Problem Statement #27: Warehouse Inventory & Pallet Location Tracker

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Architecture Selection

We chose Layered Architecture for the Warehouse Inventory & Pallet Location Tracker System.

The Warehouse Inventory & Pallet Location Tracker is designed to help warehouses manage inventory in a smarter and more organized way. It keeps track of pallet locations in 3D storage racks, checks whether a rack can safely hold the weight of a pallet before storing it, and records every stock movement using barcode or RFID scanning. We selected Layered Architecture because it divides the system into different layers, where each layer performs a specific job. This makes the system easier to understand, develop, maintain, and manage as the warehouse grows.

1. Architectural Choice

Selected Architectural Style: Layered Architecture

The Layered Architecture organizes the warehouse system into three main layers, each with a clear responsibility.

- Presentation Layer: This is the user interface used by warehouse operators and logistics supervisors to manage inventory, scan pallets, and view warehouse information.
- Business Logic Layer: This layer processes all the important warehouse operations such as inventory updates, pallet location assignment, weight validation, and barcode/RFID scanning.
- Data Layer: This layer stores all warehouse data, including pallet coordinates, SKU details, rack information, inventory records, and stock movement history.

By separating these responsibilities, the system becomes more structured and each layer can work independently while communicating through well-defined interfaces.

2. Two Reasons for Choosing Layered Architecture

Reason 1 – Better Organization of Warehouse Operations

Warehouse management involves multiple tasks happening together, such as scanning pallets, checking rack capacity, updating inventory, and assigning storage locations. With Layered Architecture, these tasks are handled by different layers instead of mixing everything into one module. This makes the system easier to understand and helps developers manage each function without affecting the others.

Reason 2 – Easy Maintenance and Future Improvements

Warehouse systems often need updates, such as adding new storage racks, supporting new scanning devices, or introducing additional inventory features. Since the business logic is separated from the user interface and database, these changes can be implemented without making major modifications to the entire system. This makes the application easier to maintain and expand in the future.

3. Security Advantage

One important advantage of Layered Architecture is that it validates warehouse operations before any information is stored in the database. Whenever a pallet is scanned or assigned to a rack, the system first checks whether the rack has enough weight capacity and whether the inventory details are valid.

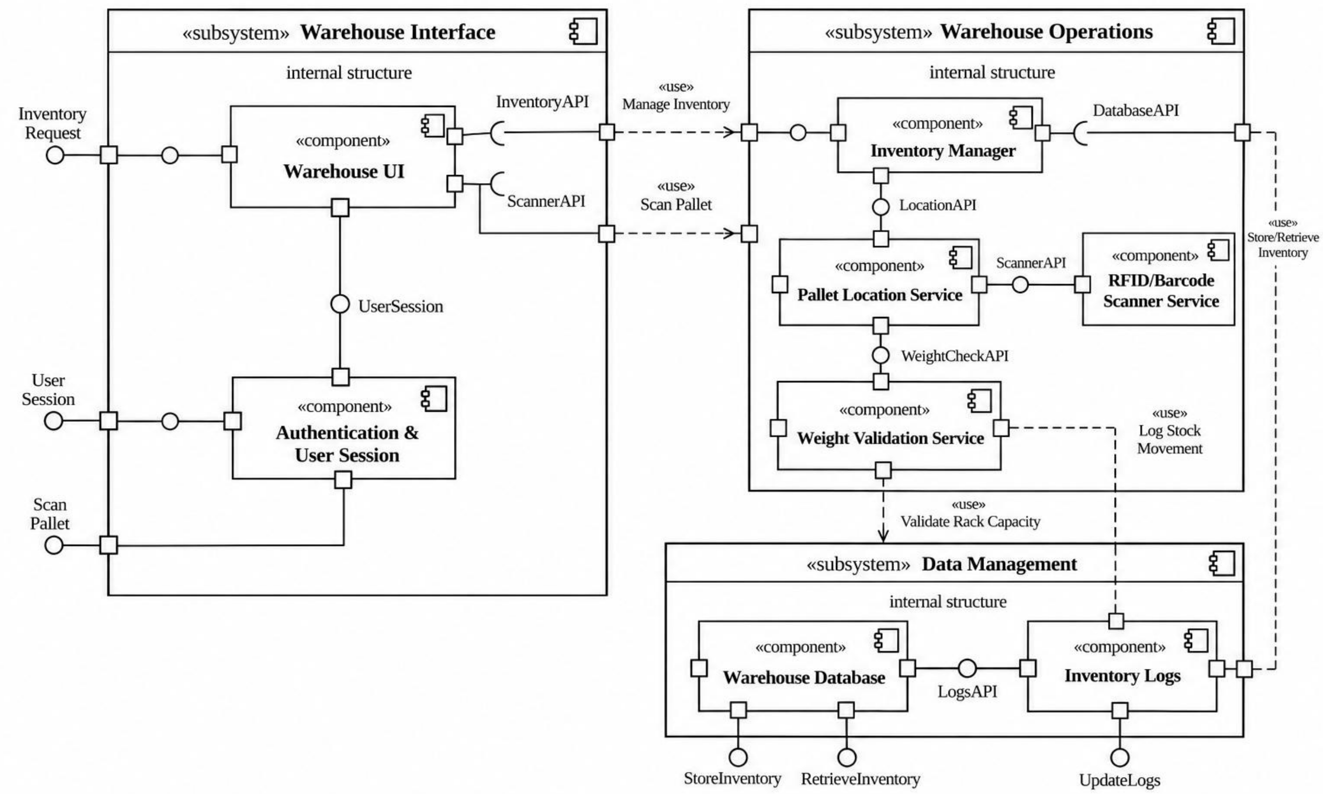
This approach helps prevent incorrect pallet placement, unauthorized inventory updates, and invalid barcode or RFID entries. As a result, only verified and accurate warehouse transactions are saved, improving the reliability of inventory records.

4. Performance Benefit

Layered Architecture also improves the overall performance of the warehouse system by assigning different responsibilities to dedicated components. The Inventory Manager handles inventory operations efficiently, while the Pallet Location Service quickly identifies available storage locations and validates rack capacity before storage.

Because each layer focuses on its own task, inventory lookups, barcode/RFID scanning, and pallet location updates can be processed faster. This allows warehouse operators to complete stock movements with minimal delay, even when managing a large number of inventory items.

Component Diagram



UML Component Diagram
Warehouse Inventory & Pallet Location Tracker

Conclusion

The Layered Architecture is the most suitable choice for the Warehouse Inventory & Pallet Location Tracker System because it provides a clean and well-organized structure for managing warehouse operations. It improves maintainability, enhances security through proper validation, and supports efficient inventory processing. By separating the user interface, business logic, and data management into different layers, the system becomes easier to develop, test, and upgrade while meeting the requirements of the warehouse management project.